

CSS Orientation Media Rule

Complete Guide: From Basics to Advanced

CSS Guide

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Chapter 1

Introduction

1.1 What is the Orientation Media Rule?

The **orientation** media feature in CSS allows you to apply different styles based on whether a device is in **portrait** or **landscape** mode. This is essential for creating truly responsive designs that adapt not just to screen size, but to how the device is being held or oriented.

```
/* Basic syntax */
@media (orientation: portrait) {
    /* Styles for portrait orientation */
}

@media (orientation: landscape) {
    /* Styles for landscape orientation */
}
```

The orientation media rule is one of the fundamental tools for mobile-first responsive design, allowing you to create interfaces that feel natural regardless of how users hold their devices.

1.2 Understanding Device Orientation

1.2.1 Portrait vs Landscape

Aspect	Portrait	Landscape	Trigger
Height/Width	Taller than wide	Wider than tall	Viewport dimensions
Aspect Ratio	< 1 (e.g., 9:16)	> 1 (e.g., 16:9)	height > width or width > height
Common Device	Mobile phone	Tablet rotated	Device sensor
Height (h)	Greater	Lesser	Device rotation
Width (w)	Lesser	Greater	Device rotation
Use Cases	Reading, messaging	Gaming, video	User preference

1.2.2 How Orientation is Determined

The browser automatically determines orientation by comparing viewport dimensions:

- **Portrait:** When `viewport-height > viewport-width`

- **Landscape:** When `viewport-width > viewport-height`
- **Square:** Treated as landscape if exactly equal

1.3 Why Use Orientation Media Queries?

- **Mobile Adaptation:** Optimize UI for different holding positions
- **Better UX:** Layouts that feel natural in each orientation
- **Device Features:** Take advantage of screen orientation capabilities
- **Game Development:** Create games playable in both orientations
- **Data Visualization:** Show different chart layouts per orientation
- **Performance:** Load resources appropriate for current orientation

Chapter 2

Core Concepts

2.1 Basic Orientation Values

2.1.1 Portrait Orientation

```
@media (orientation: portrait) {  
    /*  
        Applied when:  
        - Viewport height > Viewport width  
        - Phone held vertically  
        - Tablet in default position  
    */  
    body {  
        background: #f5f5f5;  
    }  
}
```

Common on:

- Mobile phones in default position
- Tablets held vertically
- Desktop windows taller than wide

2.1.2 Landscape Orientation

```
@media (orientation: landscape) {  
    /*  
        Applied when:  
        - Viewport width > Viewport height  
        - Phone held horizontally  
        - Tablet rotated 90 degrees  
    */  
    body {  
        background: #ffffff;  
    }  
}
```

Common on:

- Mobile phones rotated horizontally
- Tablets in landscape mode
- Desktop monitors wider than tall
- Gaming devices

2.2 Media Query Syntax

2.2.1 Simple Orientation Check

```
/* Portrait styles */
@media (orientation: portrait) {
    .container { width: 100%; }
}

/* Landscape styles */
@media (orientation: landscape) {
    .container { width: 50%; }
}
```

2.2.2 Combined with Other Media Features

```
/* Portrait on mobile phones */
@media (orientation: portrait) and (max-width: 768px) {
    body { font-size: 14px; }
}

/* Landscape on tablets */
@media (orientation: landscape) and (min-width: 768px) {
    body { display: grid; }
}

/* Not landscape */
@media not (orientation: landscape) {
    .wide-layout { display: none; }
}

/* Multiple conditions */
@media (orientation: portrait), (max-width: 600px) {
    nav { flex-direction: column; }
}
```

Chapter 3

Practical Examples

3.1 Example 1: Responsive Navigation

3.1.1 Portrait Navigation

```
/* Portrait: Vertical navigation */
@media (orientation: portrait) {
  nav {
    flex-direction: column;
    width: 100%;
  }

  nav ul {
    display: flex;
    flex-direction: column;
    gap: 10px;
  }

  nav a {
    display: block;
    padding: 12px;
    width: 100%;
    text-align: center;
  }

  .logo {
    width: 100%;
    text-align: center;
    margin-bottom: 20px;
  }
}
```

3.1.2 Landscape Navigation

```
/* Landscape: Horizontal navigation */
@media (orientation: landscape) {
  nav {
    flex-direction: row;
```



```
        display: flex;
        align-items: center;
        justify-content: space-between;
        padding: 10px 20px;
    }

    nav ul {
        display: flex;
        flex-direction: row;
        gap: 20px;
    }

    nav a {
        display: inline-block;
        padding: 8px 16px;
    }

    .logo {
        width: auto;
        text-align: left;
    }
}
```

3.2 Example 2: Mobile Game Layout

```
/* Default portrait game layout */
.game-container {
    width: 100%;
    height: 100vh;
    display: flex;
    flex-direction: column;
}

.game-board {
    flex: 1;
    background: #333;
}

.game-controls {
    height: 120px;
    background: #222;
    display: flex;
    gap: 10px;
    padding: 10px;
    justify-content: center;
}

/* Portrait: Controls at bottom */
@media (orientation: portrait) {
    .game-container {
        flex-direction: column;
    }
}
```

```
.game-board {
  flex: 1;
  min-height: 60%;
}

.game-controls {
  height: 120px;
  flex-direction: row;
  justify-content: center;
}

.game-info {
  display: flex;
  gap: 20px;
  padding: 10px;
  background: #444;
  font-size: 14px;
}

/* Landscape: Controls on side */
@media (orientation: landscape) {
  .game-container {
    flex-direction: row;
  }

  .game-board {
    flex: 1;
    width: 70%;
  }

  .game-controls {
    flex-direction: column;
    width: 30%;
    height: auto;
  }

  .game-info {
    display: flex;
    flex-direction: column;
    gap: 10px;
    padding: 10px;
  }
}
```

3.3 Example 3: Image Gallery

```
.gallery {
  display: grid;
  gap: 10px;
  padding: 10px;
}
```

```
}

/* Portrait: Single or 2-column layout */
@media (orientation: portrait) {
  .gallery {
    grid-template-columns: repeat(2, 1fr);
  }

  .gallery-item {
    aspect-ratio: 1;
  }

  .gallery-item.featured {
    grid-column: span 2;
    aspect-ratio: 2;
  }
}

/* Landscape: Multiple columns */
@media (orientation: landscape) {
  .gallery {
    grid-template-columns: repeat(4, 1fr);
  }

  .gallery-item {
    aspect-ratio: auto;
    height: 200px;
  }

  .gallery-item.featured {
    grid-column: span 2;
    grid-row: span 2;
    height: auto;
  }
}
```

3.4 Example 4: Form Layout

3.4.1 Portrait Form

```
/* Portrait: Stacked form fields */
@media (orientation: portrait) {
  form {
    max-width: 100%;
    padding: 20px;
  }

  .form-group {
    margin-bottom: 20px;
  }

  label {
```

```
        display: block;
        margin-bottom: 5px;
        font-weight: bold;
    }

    input, textarea, select {
        width: 100%;
        padding: 10px;
        font-size: 16px; /* Prevents zoom on input focus */
    }

    .form-row {
        display: block;
    }

    button {
        width: 100%;
        padding: 12px;
        margin-top: 20px;
    }
}
```

3.4.2 Landscape Form

```
/* Landscape: Side-by-side fields */
@media (orientation: landscape) {
    form {
        max-width: 800px;
        margin: 0 auto;
        padding: 15px;
    }

    .form-row {
        display: grid;
        grid-template-columns: 1fr 1fr;
        gap: 20px;
    }

    .form-row.full {
        grid-column: span 2;
    }

    label {
        margin-bottom: 8px;
    }

    input, textarea, select {
        font-size: 14px;
    }

    button {
        width: auto;
        padding: 10px 30px;
    }
}
```

```
}  
}
```

3.5 Example 5: Video Player

```
.video-container {  
  width: 100%;  
  background: #000;  
}  
  
video {  
  width: 100%;  
  height: auto;  
  display: block;  
}  
  
.video-controls {  
  background: #333;  
  color: white;  
  padding: 15px;  
}  
  
/* Portrait: Full width video */  
@media (orientation: portrait) {  
  .video-container {  
    width: 100%;  
    max-height: 50vh;  
  }  
  
  .video-info {  
    padding: 15px;  
    font-size: 14px;  
  }  
  
  .related-videos {  
    display: grid;  
    grid-template-columns: repeat(2, 1fr);  
    gap: 10px;  
    padding: 15px;  
  }  
}  
  
/* Landscape: Optimized for fullscreen */  
@media (orientation: landscape) {  
  .video-container {  
    width: 100%;  
    height: 100vh;  
  }  
  
  .video-info {  
    display: none;  
  }  
}
```

```
.related-videos {  
    display: none;  
}  
  
.video-controls {  
    position: absolute;  
    bottom: 0;  
    width: 100%;  
}  
}
```

Chapter 4

Advanced Techniques

4.1 Combining Orientation with Size Queries

4.1.1 Responsive + Orientation

```
/* Smallest phones in portrait */
@media (orientation: portrait) and (max-width: 480px) {
  body { font-size: 14px; }
  .container { padding: 10px; }
}

/* Tablets in portrait */
@media (orientation: portrait) and (min-width: 768px) {
  body { font-size: 16px; }
  .container { padding: 20px; }
}

/* Tablets in landscape */
@media (orientation: landscape) and (min-width: 768px) {
  body { font-size: 16px; }
  .sidebar { width: 250px; }
}

/* Large desktop in landscape */
@media (orientation: landscape) and (min-width: 1920px) {
  body { font-size: 18px; }
  .container { max-width: 1400px; }
}
```

4.2 Device Orientation API Integration

4.2.1 Using JavaScript with Orientation Media Queries

```
/* CSS responds to orientation changes */
@media (orientation: portrait) {
  .orientation-info::before {
    content: 'Portrait Mode';
  }
}
```

```
    }  
  }  
  
  @media (orientation: landscape) {  
    .orientation-info::before {  
      content: 'Landscape Mode';  
    }  
  }  
}
```

4.2.2 JavaScript Listeners

```
/* Add data-orientation attribute for JavaScript targeting */  
[data-orientation="portrait"] .mobile-menu {  
  position: fixed;  
  bottom: 0;  
  width: 100%;  
}  
  
[data-orientation="landscape"] .mobile-menu {  
  position: static;  
  width: auto;  
}
```

4.3 Animations Based on Orientation

```
/* Transition animations on orientation change */  
.layout {  
  transition: all 0.3s ease;  
}  
  
@media (orientation: portrait) {  
  .layout {  
    flex-direction: column;  
    gap: 10px;  
  }  
}  
  
@media (orientation: landscape) {  
  .layout {  
    flex-direction: row;  
    gap: 20px;  
  }  
}  
  
/* Animation on orientation change */  
@media (orientation: landscape) {  
  .sidebar {  
    animation: slideIn 0.3s ease;  
  }  
}
```



```
@keyframes slideIn {  
  from {  
    transform: translateX(-100%);  
  }  
  to {  
    transform: translateX(0);  
  }  
}
```

Chapter 5

Real-World Use Cases

5.1 Use Case 1: News/Article App

```
/* Portrait: Single column */
@media (orientation: portrait) {
  .article-layout {
    display: block;
  }

  .article-content {
    width: 100%;
  }

  .sidebar {
    display: none;
  }

  img {
    width: 100%;
    height: auto;
  }
}

/* Landscape: Content + Sidebar */
@media (orientation: landscape) {
  .article-layout {
    display: grid;
    grid-template-columns: 1fr 300px;
    gap: 20px;
  }

  .article-content {
    width: auto;
  }

  .sidebar {
    display: block;
  }
}
```

5.2 Use Case 2: Dashboard Application

```
/* Portrait: Stacked widgets */
@media (orientation: portrait) {
  .dashboard {
    display: flex;
    flex-direction: column;
  }

  .widget {
    width: 100%;
    min-height: 200px;
    margin-bottom: 15px;
  }

  .chart {
    height: 250px;
  }
}

/* Landscape: Grid layout */
@media (orientation: landscape) {
  .dashboard {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 15px;
  }

  .widget {
    width: auto;
  }

  .widget.full-width {
    grid-column: span 3;
  }

  .widget.half-width {
    grid-column: span 1;
  }

  .chart {
    height: 300px;
  }
}
```

5.3 Use Case 3: Shopping Cart

```
/* Portrait: Single column cart */
@media (orientation: portrait) {
  .cart-container {
    display: block;
  }
}
```

```
}

.cart-items {
  width: 100%;
  margin-bottom: 20px;
}

.cart-summary {
  width: 100%;
  position: sticky;
  bottom: 0;
  background: white;
  border-top: 1px solid #ddd;
  padding: 15px;
}

.item-row {
  display: flex;
  gap: 10px;
  padding: 15px 0;
  border-bottom: 1px solid #eee;
}
}

/* Landscape: Two column layout */
@media (orientation: landscape) {
  .cart-container {
    display: grid;
    grid-template-columns: 1fr 300px;
    gap: 20px;
  }

  .cart-items {
    width: auto;
  }

  .cart-summary {
    position: static;
    border: 1px solid #ddd;
    border-radius: 8px;
    padding: 20px;
  }
}
```

Chapter 6

Browser Support

6.1 Current Support Status

The orientation media rule has excellent browser support:

Browser	Version	Status
Chrome/Edge	All versions	Full Support
Firefox	All versions	Full Support
Safari	All versions	Full Support
Opera	All versions	Full Support
Internet Explorer	9+	Full Support
Mobile Browsers	All versions	Full Support

6.2 Fallback Strategy

Since orientation media queries are universally supported, fallbacks are rarely needed. However, for basic support:

```
/* Default styles work on all browsers */
.container {
  width: 100%;
  padding: 20px;
}

/* Enhanced with orientation support */
@media (orientation: landscape) {
  .container {
    display: grid;
    grid-template-columns: 2fr 1fr;
  }
}
```

Chapter 7

Best Practices

7.1 Key Recommendations

7.1.1 1. Always Test Both Orientations

Test your design in:

- Actual mobile devices
- Browser DevTools device simulator
- Physical device rotation
- Both portrait and landscape

7.1.2 2. Avoid Horizontal Scrolling

```
/* Bad: Forces horizontal scroll */
@media (orientation: landscape) {
  .wide-content {
    width: 200%; /* Too wide! */
  }
}

/* Good: Fits within viewport */
@media (orientation: landscape) {
  .wide-content {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
  }
}
```

7.1.3 3. Optimize for Limited Landscape Height

Landscape mode on phones has limited height. Account for this:

```
@media (orientation: landscape) {
  /* Reduce padding/margins */
  body { padding: 5px; }
```

```
/* Make text smaller if needed */
body { font-size: 14px; }

/* Stack multiple elements */
nav { display: flex; }
}
```

7.1.4 4. Font Size Considerations

```
/* Portrait: Normal text sizes */
@media (orientation: portrait) {
  body { font-size: 16px; }
  h1 { font-size: 28px; }
}

/* Landscape: Slightly smaller */
@media (orientation: landscape) {
  body { font-size: 14px; }
  h1 { font-size: 24px; }
}
```

7.1.5 5. Input Field Sizing

Always ensure `font-size: 16px` on inputs in portrait to prevent zoom:

```
@media (orientation: portrait) {
  input, textarea, select {
    font-size: 16px; /* Prevents autozoom */
  }
}

@media (orientation: landscape) {
  input, textarea, select {
    font-size: 14px; /* Can be smaller */
  }
}
```

Chapter 8

Common Issues & Solutions

8.1 Issue 1: Content Doesn't Adapt on Rotation

Problem: Styles don't update when device rotates.

Causes:

- Missing media queries
- Viewport meta tag not configured
- CSS not compiled/refreshed

Solution:

```
/* Ensure viewport meta tag is present */
<meta name="viewport" content="width=device-width,
                                initial-scale=1.0">

/* Provide orientation styles */
@media (orientation: portrait) {
    .layout { flex-direction: column; }
}

@media (orientation: landscape) {
    .layout { flex-direction: row; }
}
```

8.2 Issue 2: Text Too Small in Landscape

Problem: Text becomes unreadable when rotated.

Solution:

```
/* Don't go below 12px for body text */
@media (orientation: landscape) {
    body {
        font-size: 14px; /* Not 10px! */
    }
}
```



```
/* Use CSS clamp for automatic scaling */
@media (orientation: landscape) {
  body {
    font-size: clamp(12px, 1.5vw, 16px);
  }
}
```

8.3 Issue 3: Fixed Elements Breaking Layout

Problem: Fixed positioned elements cause issues in landscape.

Solution:

```
/* Portrait: Fixed bottom navbar */
@media (orientation: portrait) {
  .navbar {
    position: fixed;
    bottom: 0;
    width: 100%;
    height: 60px;
  }

  body {
    padding-bottom: 60px;
  }
}

/* Landscape: Static navbar */
@media (orientation: landscape) {
  .navbar {
    position: static;
    height: auto;
  }

  body {
    padding-bottom: 0;
  }
}
```

8.4 Issue 4: Images Not Scaling Properly

Problem: Images look distorted or don't fit in landscape.

Solution:

```
@media (orientation: portrait) {
  img {
    width: 100%;
    height: auto;
  }
}

@media (orientation: landscape) {
```

```
img {  
  max-width: 100%;  
  height: auto;  
  max-height: 80vh;  
}
```

Chapter 9

Summary & Quick Reference

9.1 When to Use Orientation Queries

Use orientation media queries when:

- Building mobile-first responsive sites
- Creating apps that support device rotation
- Optimizing games for both orientations
- Building dashboards and data displays
- Creating mobile applications

9.2 When NOT to Use (Use Size Queries Instead)

- Desktop-only sites (rarely rotate)
- When screen width/height is the main concern
- Simple responsive layouts (use max-width instead)

9.3 Quick Reference

9.3.1 Basic Syntax

```
/* Portrait orientation */
@media (orientation: portrait) {
  /* Your styles */
}

/* Landscape orientation */
@media (orientation: landscape) {
  /* Your styles */
}

/* Combined with other media features */
```

```
@media (orientation: portrait) and (max-width: 768px) {  
    /* Styles for portrait phones */  
}  
  
/* Not operator */  
@media not (orientation: portrait) {  
    /* Landscape styles */  
}
```

9.3.2 Common Patterns

Pattern	Use Case
Portrait only	Single-column layout
Landscape only	Side-by-side layout
Portrait + size	Mobile-specific
Landscape + size	Tablet/desktop
Multiple orientation checks	Complex apps

9.3.3 Complete Starter Template

```
/* Base styles */  
body {  
    font-family: sans-serif;  
    margin: 0;  
    padding: 0;  
}  
  
/* Portrait orientation */  
@media (orientation: portrait) {  
    body {  
        font-size: 16px;  
    }  
  
    .container {  
        padding: 20px;  
        display: flex;  
        flex-direction: column;  
    }  
  
    .sidebar {  
        display: none;  
    }  
  
    .main-content {  
        width: 100%;  
    }  
}  
  
/* Landscape orientation */  
@media (orientation: landscape) {  
    body {
```

```
        font-size: 14px;
    }

    .container {
        padding: 15px;
        display: grid;
        grid-template-columns: 1fr 250px;
        gap: 20px;
    }

    .sidebar {
        display: block;
    }

    .main-content {
        width: auto;
    }
}
```

Thank you for reading!

For more information, visit [MDN Web Docs](#) and [web.dev](#)